

Harsh Giri

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EXPERIENCE

ML Apprentice | Microsoft SAP Edunet

Jan 2025 – Mar 2025

- Engineered a **multi-disease clinical prediction system** on **45,000+ patient records**, ensembling **Random Forest, SVM**, and **Neural Networks** to boost accuracy from **82% → 93.7%**, achieve **0.96 AUC**, and cut adverse-event false negatives by **18%**.
- Automated feature extraction via **PCA** and **10-fold cross-validation**, implementing **SMOTE** to improve high-risk minority-class recall by **23%**, critical for rare drug-interaction detection in **clinical decision support** systems.
- Optimized **ML training pipelines** with drug co-administration pattern encoding and clinical feature engineering, achieving **3× faster** experimentation cycles via parallelized cross-validation, reducing model deployment time by **40%**.

Open-Source Contributor | Apache OpenOffice / GirlScript Summer of Code

2024 – 2026

- Contributed **5+ production bug fixes** across **C++, Java**, and **XML** in Apache OpenOffice, leveraging **AST-based static code analysis** across **15+ modules**, improving codebase maintainability by **30%**.
- Implemented **CI/CD regression testing** workflows integrated with the Apache build system, reducing post-merge defect rates by **25%** across heterogeneous C++/Java codebases.
- Accelerated developer onboarding by authoring **Doxygen documentation** for **15+ modules**, cutting contributor ramp-up time by an estimated **35%**.

PROJECTS

Enterprise Grade Agentic RAG | Guardrailed Q&A over Technical Documentation | LangGraph, Qdrant, NeMo Guardrails, FastAPI

[Github](#)

- Engineered a **guardrailed agentic RAG service** on **FastAPI + LangGraph**, routing each turn through a **typed-intent state machine** with conditional edges, LLM query rewriting for multi-turn coreference, and a **thread-scoped checkpoint**; a **fail-closed NeMo Guardrails (Colang) gate** on **Llama-3.1-8B** blocks jailbreaks and off-topic input before any embedding, vector search, or 70B call.
- Architected **two-stage retrieval Qdrant cosine ANN top-15 → quantised ONNX cross-encoder rerank → top-5** — over a **64-document corpus seeded with 57 distractor files**, behind a **dual-backend abstraction** swapping **3072-dim Gemini ↔ 768-dim MPNet** embeddings and **Qdrant Cloud ↔ embedded index** from a single config flag.
- Deployed on **Cloud Run** (non-root container, API-key auth, CORS allowlist) with a **Portkey gateway** for **70B→8B fallback**, response caching and retry-on-429/503; instrumented with **Logfire/LangSmith tracing** and a **RAGAS suite** (faithfulness, context precision/recall, answer correctness) gated by **GitHub Actions CI running 82 unit tests**.

AI-Powered Game Recommendation Engine | KNN, Cosine Similarity, FastAPI, HNSW

[Github](#) / [Demo](#)

- Engineered a **hybrid recommendation engine** combining **KNN** and **cosine similarity** over **50,000+ game records**, with **HNSW indexing** achieving **<35ms query latency** and **92% retrieval precision**.
- Deployed a **production-grade FastAPI backend** sustaining **1,000+ daily requests** at **99.5% SLA**, integrating a **Redis caching layer** that reduced database load by **40%** during peak traffic.
- Orchestrated end-to-end **MLOps pipeline** with **async REST API** endpoints, cutting cold-start inference time by **30%** and improving system throughput by **3×** under concurrent load.

Text Extraction with YOLOv11 and OCR | YOLOv11, Tesseract, OpenCV

[Github](#) / [Demo](#)

- Engineered a **two-stage document intelligence pipeline** fusing **YOLOv11** detection with **Tesseract OCR**, fine-tuned on **8,000+ annotated images** to achieve **mAP@0.5 of 91.3%** at **~95ms/image**.
- Architected an **OpenCV preprocessing pipeline** (grayscale, adaptive thresholding, binarization) boosting OCR accuracy by **28%**, enabling **650+ docs/min** with structured **JSON output** for downstream ETL.
- Optimized **YOLO fine-tuning** via **mixed-precision training** and **transfer learning**, cutting training convergence by **35%** while maintaining **>90% precision** on unseen document layouts.

TECHNICAL SKILLS

Languages & Tools: Python, C++, Java, SQL, Bash, Git, Docker, Jupyter

LLM, Agentic Systems & Guardrails: LangGraph, LangChain, Agentic Workflows, Tool Calling, Multi-Turn State Management, Conversational Memory & Checkpointing, Prompt Engineering, Structured Output, Context Window Optimization, Tokenization, NeMo Guardrails (Colang), Prompt-Injection & Jailbreak Defense, LLM Gateways (Portkey), Model Fallback & Caching

RAG, Retrieval & Evaluation: Qdrant, FAISS, ChromaDB, HNSW / ANN Indexing, Cross-Encoder Reranking (ONNX), Two-Stage Retrieval, Semantic Search, Chunking Strategies, Sentence Transformers, OpenAI / Gemini / HuggingFace Embeddings, RAGAS (Faithfulness, Context Precision / Recall, Answer Correctness), LLM-as-Judge, Hallucination Detection, Logfire, LangSmith, Distributed Tracing

ML, Backend & MLOps: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Ensemble Methods (Random Forest, SVM, Neural Networks), CNN Architectures (ResNet, VGG), Transfer Learning, Mixed-Precision Training, PCA, SMOTE, K-Fold Cross-Validation, FastAPI, Async REST APIs, Pydantic, Streamlit, Redis Caching, Google Cloud Run, GitHub Actions CI/CD, pytest

CERTIFICATIONS

Oracle Certified Data Science Professional — Oracle University

[Link](#)

Oracle Certified AI Foundations Associate — Oracle University

[Link](#)

EDUCATION

Dr. APJ Abdul Kalam Technical University (B.Tech)

Jun 2024 – May 2028

B.Tech in Computer Science (Data Science) | Ghaziabad, Uttar Pradesh